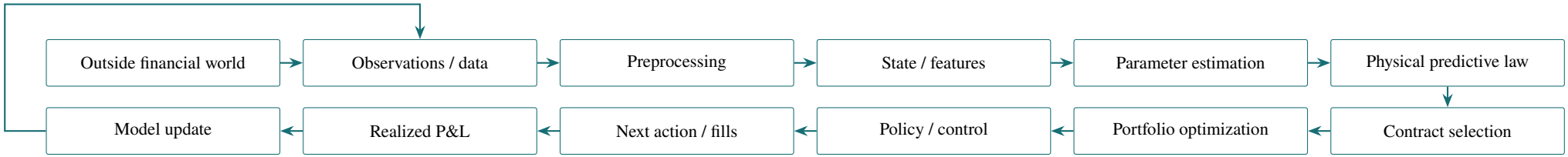
Multi-asset futures: data, thesis and path law

One-period baseline retained; optional slow thesis state and multi-period control use the same gain and cash maps.



Observations update beliefs; only decision dates (or mandatory risk events) trigger strategic orders. Execution has its own faster clock.

Three clocks. Observations arrive at o_j; decisions occur at t_k with spacing Δ_d; the economic thesis expires at a fixed T^{th} (e.g. weeks). Planning depth H counts decision intervals, $t_{k+H} \le T^{\text{th}}$; it is not the observation count. Below k indexes decisions and $\mathcal{F}_k$ contains only released information. Freeze training length N, PCA rank $r < n$, scales, priors, calendar/surprise laws, risk/cost bounds, seed and validation splits. All money uses one currency. Do not roll the thesis deadline forward after losses.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
F0. Data, gains, coordinates $O_{\le k}$: dated records; $g_s, x_s \in \mathbb{R}^n$. $D \in \mathbb{R}^{n\times n}$ positive diagonal; $\bar r \in \mathbb{R}^n$. $B = [V\ U] \in \mathbb{R}^{n\times n}$ orthogonal, $V \in \mathbb{R}^{n\times r}$.	Released settlements, multipliers m_i, liquidity, funding, margin and fills enter $g_{i,s} = m_i(F_{c_i(s-1),s} - F_{c_i(s-1),s-1})$, using the maturity held over that interval. Never splice a roll price jump into gain. On N training gains, $D_{ii} = \mathrm{sd}(g_i) > 0$, $r_s = D^{-1}g_s$, $\bar r = N^{-1}\sum r_s$. Eigenvectors of $S_r = (N-1)^{-1}\sum(r_s - \bar r)(r_s - \bar r)^\top$ give B, ordered with fixed signs/ties. Fast factor/residual coordinates: $x_s = B^\top(r_s - \bar r)$. Freeze/recompute the whole block in one basis. **To F1–F3, F5.**	NumPy data `(N,n)`, basis/covariance `(n,n)`, state `(n,)`; timestamped contract records.
F1. One-step baseline fit $\widehat A \in \mathbb{R}^{n\times n}$, $\widehat Q \in \mathbb{S}^n_{++}$; $\mu_k \in \mathbb{R}^n$, $\Sigma_k \in \mathbb{S}^n_{++}$. $\widehat\theta$: all fitted coefficients.	$x_{s+1} = Ax_s + \varepsilon_{s+1}$, $\varepsilon \sim \mathcal{N}(0, Q)$. $\widehat A = (\sum x_{s+1}x_s^\top)(\sum x_s x_s^\top + \lambda I)^{-1}$, $\widehat Q = (N-1)^{-1}\sum(x_{s+1} - \widehat A x_s)(x_{s+1} - \widehat A x_s)^\top + \epsilon I$, $\lambda, \epsilon > 0$. Reject spectral radius $\rho(\widehat A) \ge r_* < 1$. Baseline $g_{k+1} \mid \mathcal{F}_k \sim \mathcal{N}(\mu_k, \Sigma_k)$, $\mu_k = D(\bar r + B\widehat A x_k)$, $\Sigma_k = DB\widehat Q B^\top D$. **To F3, F5; this is one step, not a thesis-horizon forecast.**	Arrays `(n,n)`, `(n,)`; fit/transition callables; SciPy Gaussian law.
F2. Slow thesis extension $u_k \in \mathbb{R}$: latent valuation anchor; $\ell_k \in \mathbb{R}$: observed reference-spread level. $d_k = u_k - \ell_k$: remaining gap; $y_k \in \mathbb{R}$: economic proxy. $c_{k+1} \in \mathbb{R}^p$: catalyst surprises; $b_k = \mathrm{Law}(u_k \mid \mathcal{F}_k)$.	Choose a fixed reference basket $w_0 \in \mathbb{R}^n$, capital $A_0 > 0$, and a predeclared economic valuation proxy y in the same units: $\ell_{k+1} = \ell_k + w_0^\top g_{k+1}/A_0$, $u_{k+1} = \bar u + \phi(u_k - \bar u) + \beta^\top c_{k+1} + \sigma_u\eta_{k+1}$, $y_k = u_k + \tau_y\nu_k$, $x_{k+1} = Ax_k + a_d d_k + J_c c_{k+1} + \varepsilon_{k+1}$. Here $a_d \in \mathbb{R}^n, J_c \in \mathbb{R}^{n\times p}, \beta \in \mathbb{R}^p$, $0 < \phi \le 1$, $\sigma_u, \tau_y > 0$; independent standardized Gaussian η, ν. Persistence is estimated ϕ near one and small σ_u, not immunity to evidence. At announced releases $c \sim \mathcal{N}(\bar c, C_c)$, $C_c \in \mathbb{S}^p_+$, fitted to past surprises; otherwise $c = 0$. Filter from a proper prior: $b_{k+1}(u) \propto p(y_{k+1} \mid u)\int p(x_{k+1}, u \mid x_k, u', \ell_k) b_k(u')\,\mathrm{d}u'$; maximize predictive log likelihood (EM) for coefficients. Between decision-grid VAR observations, released y's update the current anchor likelihood; quotes update risk. $a_d = J_c = 0$ recovers F1. **To F3–F6.**	Filter mean/variance `(1,)`, `(1,1)`; coefficients `(n,p)`, `(p,)`. Proxy map and catalyst calendar fixed before testing.
F3. Joint physical path law $Z_k = (x_k, u_k, \ell_k) \in \mathbb{R}^{n+2}$; K^P_θ: F1/F2 transition. $\mathsf{P}^H_k \in \mathcal{P}((\mathbb{R}^{n+2})^{H+1})$. $L_k \in \mathbb{R}^{n\times n}$, $\rho_\mu \ge 0$: mean uncertainty.	$\mathsf{P}^H_k(\mathrm{d}Z_{k:k+H}) = b_k(\mathrm{d}u_k)\prod_{j=k}^{k+H-1} K^P_{\widehat\theta}(\mathrm{d}Z_{j+1} \mid Z_j)$, observed x_k, ℓ_k fixed, catalysts integrated. The endpoint $\mathbb{P}(Z_{k+H} \mid \mathcal{F}_k)$ is only a marginal. For F1 alone it gives $x_{k+H} \sim \mathcal{N}(A^H x_k, \sum_{j=0}^{H-1} A^j Q (A^j)^\top)$; cumulative gain and interim risk need cross-time dependence. Use $\mu_k = \mathbb{E}^P_k g_{k+1}$, $\Sigma_k = \mathrm{Cov}^P_k(g_{k+1})$, including anchor uncertainty. Block-bootstrap refits give $\mu_k^{(b)}$, $L_k L_k^\top = \widehat{\mathrm{Cov}}_b(\mu_k^{(b)}) + \epsilon_\mu I$; ρ_μ is a frozen quantile of $\|L_k^{-1}(\mu_k^{(b)} - \mu_k)\|$. Hold each fitted parameter set fixed per path. **To F5–F6.**	Paths `(N_paths,H+1,n+2)`, gains `(N_paths,H,n)`; Gaussian distribution or simulation callable; bootstrap means `(N_boot,n)`.
F4. Thesis status / expression $p_k = b_k(u_k - \ell_k > \delta) \in [0,1]$; $z_k \in \{0, 1, \dagger\}$: waiting, active, closed. $p_{\text{on}} > p_{\text{off}}, \delta, \delta_{\text{exit}} \ge 0$; eligible mask $j_k \in \{0,1\}^n$.	For a positive-gap thesis, activate at $p_k \ge p_{\text{on}}$; retain it between thresholds. Close on convergence $\mathbb{E}_k d_k \le \delta_{\text{exit}}$, $p_k \le p_{\text{off}}$, released adverse evidence $y < y_{\min}$ or $c_j < \underline{c}_j$, or T^{th}; freeze thresholds and label reasons. Waiting forbids new thesis risk; † cancels entry orders and flattens, and is absorbing until a new thesis record. Risk/margin failures act immediately. The thesis is u, d, b, not a contract ID: choose maturities, spreads, hedges or zero anew. Shares/options require their own pricing/ledger. Eligibility $j_i = 1$ requires valid liquid quotes, capped spreads and dates before notice/roll cutoff; held ineligible legs exit. Disabling this extension uses F1/F5 without the thesis gate. **To F5–F6.**	Scalar probabilities, mode/deadline record, Boolean mask `(n,)`. Log thesis exit separately from temporary zero exposure.

Futures: strategic control, execution and accounting

The baseline uses H=1. Longer planning adds continuation value; observing another tick does not itself authorize another position change.

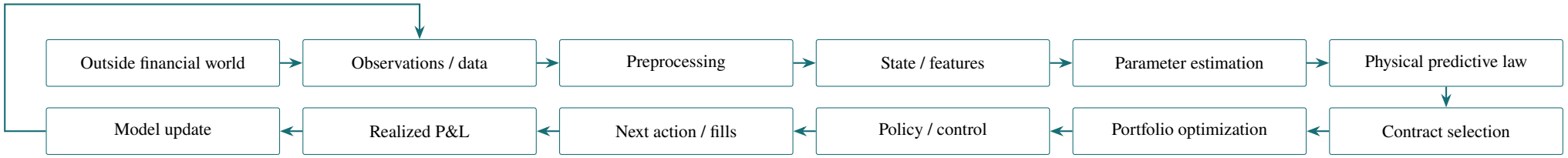
Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
F5. Baseline target / constraints $s^{\text{acct}} = (q^-, X, M) \in \mathbb{Z}^n \times \mathbb{R}^2$, $W = X + M$. Target $q \in \mathbb{Z}^n$, roll map $\mathcal{R}(q) \in \mathbb{Z}^{2n}$. $s, \eta \in \mathbb{R}^{2n}_+$, $\ell^M \in \mathbb{R}^n_+$: spread/impact/margin.	With diagonal roll flags R^{roll}, $a = \mathcal{R}(q) = (-R^{\text{roll}}q^-, q - (I - R^{\text{roll}})q^-)$. Read spreads/margin; fit nonnegative impact from fills. Define $K(a) = s^\top\lvert a\rvert + \sum \eta_i a_i^2$, $M^+(q) = (\ell^M)^\top\lvert q\rvert$, $d^c(q) = (R^M - R^f)M^+(q) - R^f K(a)$, $v(q) = q^\top \Sigma_k q$. Baseline score $J_1(q) = [q^\top\mu_k - \rho_\mu\|L_k^\top q\| + d^c(q)]/A_0 - \gamma v(q)/(2A_0^2)$. Feasible C_k: $\lvert q_i\rvert \le j_i \bar{q}_i$, $\|Dq\|_1 \le L_g$, $\lvert V^\top Dq\rvert \le b$, $\lvert H_g Dq\rvert \le c_g$, $W - K - M^+ \ge R_{\text{cash}}$, stress-equity floors, and $-q^\top\mu + \rho_\mu\|L^\top q\| - d^c + \sqrt{v}\,\varphi(\Phi^{-1}(\alpha))/(1-\alpha) \le L_{\text{ES}}$. This Gaussian ES formula applies only to the one-step Gaussian law; use simulated loss ES for longer paths. **To F6.**	CVXPY integer vector `(n,)` or bounded enumeration; constraints use supplied caps, $H_g \in \mathbb{R}^{p_g \times n}$. Gross factors $R^f, R^M > 0$; normal PDF/CDF φ, Φ.
F6. Bellman / stochastic MPC $s_k = (\mathcal{F}_k$ summary, $b_k, z_k, T^{\text{th}} - t_k, s_k^{\text{acct}}$, pending orders). $V^h, Q^h : \mathcal{S} \to \mathbb{R}$, $\mathcal{S} \times \mathcal{A} \to \mathbb{R}$. $\eta_T, h_0 \ge 0$: turnover/switch preferences.	Let $\Pi = W' - R^f W$ from F7 and $r_F = \Pi/A_0 - \gamma\,\mathrm{Var}_k(\Pi)/(2A_0^2) - \rho_\mu\|L_k^\top q\|/A_0 - \eta_T\|a\|_1$. $Q_k^h(s,a) = \mathbb{E}_k^P[r_F + V_{k+1}^{h-1}(s')]$, $V_k^h(s) = \max_{a \in \mathcal{A}_k(s)}\{Q_k^h(s,a) - h_0 \mathbf{1}_{a \ne 0}\}$. Choose $V^0 = 0$ to recover F5 at $H = 1, \eta_T = h_0 = 0$; use a specified continuation value at a shorter planning boundary, and $-K^{\text{liq}}/A_0$ at the thesis endpoint. Simulate proxy observations and re-filter at future nodes: actions never see latent states or later shocks. Stochastic MPC executes the first action of this nonanticipative tree; full Bellman control uses the remaining horizon. No-action region: $\max_{a \ne 0}[Q^H(s,a) - Q^H(s,0)] \le h_0$, if holding is feasible. F4 hysteresis and this band suppress reversals; mandatory exits/rolls override them. **To F7.**	Scenario tree / value callable; finite belief representation $\varphi_s(s) \in \mathbb{R}^{d_s}$. Costs are already in Π; turnover and switch penalties are preferences, not second cash debits.
F7. Separate execution / ledger $a^{\text{str}} \in \mathbb{Z}^{2n}$: strategic parent order; $y_e, v_e \in \mathbb{R}^{2n}$: remaining order/rate. $\Lambda_e \succ 0, \Omega_e \succeq 0 \in \mathbb{R}^{2n \times 2n}$, $T_e \ll \Delta_d$, $\lambda_e \ge 0$.	Execution chooses $v_e = -\dot{y}_e$ to minimize $\int_0^{T_e}[s_e^\top\lvert v_e\rvert + v_e^\top\Lambda_e v_e + \lambda_e y_e^\top\Omega_e y_e]\,\mathrm{d}u$, $y_e(0) = a^{\text{str}}$, $y_e(T_e) = 0$, subject to lot/rate/interim-risk limits. Fit impact Λ_e from fills and quote-change covariance Ω_e; carry unfilled quantities if completion fails. Feed expected cash cost back into F5/F6, replacing its cost estimate; execution-risk penalty is separate. Full immediate fills remain the baseline: $q^+ = q^- + \tilde{a}^{\text{old}} + \tilde{a}^{\text{new}}$, $X^+ = X + M - M^+(q^+) - K(\tilde{a})$, $W' = R^f X^+ + R^M M^+(q^+) + (q^+)^\top g'$, $\Pi = W' - R^f W$. Reset $M' = (\ell^{M'})^\top\lvert q^+\rvert$, $X' = W' - M'$. With gradual fills, accrue cash event by event and use $\int q(u)^\top\,\mathrm{diag}(m)\,\mathrm{d}F(u)$ over actual maturities. **To next account/filter.**	Execution optimizer/callable; order/fill arrays `(2*n,)`; event ledger. Strategic controller chooses exposure; execution chooses timing/size of child orders.
F8. Optional RL; when / interface $Q_\psi : \mathbb{R}^{d_s+2n} \to \mathbb{R}$: learned action value; $\bar{\psi}$: target parameters.	Input the full finite belief/account/mode/deadline encoding and parent action. Train from supported transitions (s, a, r_F, s') on $\mathcal{L}(\psi) = \mathbb{E}[(Q_\psi(s,a) - \{r_F - h_0\mathbf{1}_{a \ne 0} + \max_{a' \in \mathcal{A}(s')} Q_{\bar{\psi}}(s', a')\})^2]$, replacing the maximum by terminal value at the endpoint. Use RL when repeated large control solves need approximation, or action-dependent fills/information must be learned from adequate data. It is unnecessary when the fitted law and constrained MPC/Bellman solve are tractable. Validate simulator/offline coverage and preserve hard constraints; RL cannot supply an unidentified thesis edge. **Replaces F6 value solve only.**	Model callable / tensors; sufficient belief encoding (not an arbitrary posterior mean), feasible-action mask, replay transitions. Unseen actions require evidence, not extrapolation claims.
F9. Validation / gates $d_j^{\text{test}} = (\Pi_j - \Pi_j^{\text{bench}})/A_0$; J test decisions; $\widehat{\Gamma}_l$: sample lag covariance.	Chronological train/select/test blocks; test joint-path calibration, thesis persistence, catalyst forecasts, turnover, execution and net gains against frozen benchmarks. $Z = \sqrt{J}(\bar{d}^{\text{test}} - \delta_{\text{econ}})/\sqrt{\widehat{\Omega}}$, $\widehat{\Omega} = \widehat{\Gamma}_0 + 2\sum_{l=1}^{L_v}(1 - l/(L_v+1))\widehat{\Gamma}_l$. Predeclare thresholds/multiplicity; resample blocks preserving overlap and longer observed dependence. Gate solves only for valid inputs and nonempty feasible sets; otherwise funded liquidation or default/stop. Save posterior, gap, value difference and gate reason. **To next F0 and action gate.**	Scores `(J,)`, path diagnostics, block resampling; mode/risk/model/execution exits have distinct reason codes.

Composition: $O_{\le k} \to (x_k, \ell_k, y_k) \to (\widehat{\theta}, b_k, d_k, z_k) \to \mathsf{P}_k^H \to (Q_k^H, \mathcal{A}_k) \to a_k^{\text{str}} \to$ child fills $\to W_{k+1} \to b_{k+1}$.

$$a_k^* = F(O_{\le k}, s_k^{\text{acct}}, \widehat{\theta}) = \mathcal{E}_{\text{F7}}\left[\text{Gate}_{\text{F4,F9}}\left(\arg\max_{a \in \mathcal{A}_k(s_k)}\left\{\mathbb{E}_{\mathsf{P}_k^H}[r_F(s_k, a, s') + V_{k+1}^{H-1}(s')] - h_0\mathbf{1}_{a \ne 0}\right\}\right)\right].$$

SPY options: thesis paths under P, option values under Q

Baseline: one-period Heston/utility MPC. Extension: a slow valuation anchor, a fixed economic deadline and adaptive expression.



Keep the physical thesis separate from its current instrument. Observations update beliefs; strategic decisions and child-order execution use separate clocks.

Three clocks. Observations at o_j may be intraday; strategic decisions at t_k may be daily; the economic horizon ends at a fixed T^{th}, e.g. four weeks. H is the number of planned decision intervals, with $t_{k+H} \le T^{\text{th}}$. Released information is $\mathcal{F}_k$. Freeze priors, proxy definitions, calendar/surprise laws, fits, bounds, simulation count P, seed, numerical grids and test blocks. Between decisions observe/filter; only scheduled decisions or mandatory risk/exercise events authorize portfolio changes.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
O0. Released data / features $S, Y^v > 0$; J listed legs with strike K_j, expiry T_j, multiplier m_j, sign $\chi_j = \pm 1$. $C, \omega \in \mathbb{R}^J$, $X \in \mathbb{R}^{2+2J}$; economic proxy $y \in \mathbb{R}$.	Synchronize spot, bid/ask, dividends, funding, liquidity and notices; reject stale/crossed quotes. For an interval of Δ years set $Y^v = \max\{\epsilon_v, \sum_l (\Delta_l \log S)^2/\Delta\}$. Dollar midpoint $C_j = m_j(\text{bid}_j + \text{ask}_j)/2$; $\omega_j = \max\{\omega_{\min}, m_j(\text{ask}_j - \text{bid}_j)/2\}$, positive floors. $X = (S, \log Y^v, C, \omega)$, plus mask/calendar. Map released economic evidence to a predeclared log-valuation proxy y; record its formula and timestamps. **To O1–O4.**	Records; NumPy `(J,)`. Optional option grid `(N_K,N_T,2)` plus mask. Economic evidence is data, not a retrospectively revised story.
O1. Fast market / slow thesis $Z = (S, v, u) \in \mathbb{R}^2_+ \times \mathbb{R}$; u: latent log-valuation anchor; $d = u - \log S$: convergence gap. $c \in \mathbb{R}^P$: catalyst surprises. $\theta \in \Theta \subset \mathbb{R}^{12+P}$: seven Heston parameters plus slow coefficients.	Between decision dates, freeze u_k and use $\mathrm{d}S/S = \mu_k^P(S, u_k)\,\mathrm{d}t + \sqrt{v}\,\mathrm{d}W_1^P$, $\mathrm{d}v = \kappa_P(\bar{v}_P - v)\,\mathrm{d}t + \xi\sqrt{v}(\rho\,\mathrm{d}W_1^P + \sqrt{1-\rho^2}\,\mathrm{d}W_2^P)$, $\mu_k^P = \mu + \kappa_d(u_k - \log S)$. Slow model: $u_{k+1} = \bar{u} + \phi(u_k - \bar{u}) + \beta^\top c_{k+1} + \sigma_u \eta$, $y = u + \tau_y \nu$, $\beta \in \mathbb{R}^P$. Independent $\eta, \nu \sim \mathcal{N}(0,1)$; fit $0 < \phi \le 1$, $\sigma_u, \tau_y > 0$. At announced releases $c \sim \mathcal{N}(\bar{c}, C_c)$, fitted to past surprises; otherwise $c = 0$. Under $\mathbb{Q}$, stock drift is r, variance drift $\kappa_Q(\bar{v}_Q - v)$; ξ, ρ are shared. Risk prices: $\lambda_1 = (\mu_k^P - r)/\sqrt{v}$, $\xi\sqrt{v}(\rho\lambda_1 + \sqrt{1-\rho^2}\lambda_2) = \kappa_P(\bar{v}_P - v) - \kappa_Q(\bar{v}_Q - v)$. Impose $2\kappa_P\bar{v}_P, 2\kappa_Q\bar{v}_Q \ge \xi^2$, $\lvert\rho\rvert < 1$, positive variance coefficients and admissible moment/martingale bounds. Apply announced cash-dividend jumps on the stated positive-price domain. $\kappa_d = 0$ recovers constant drift. **To O2–O3.**	Fast `(2,)` / slow scalar; parameter vector `(12+p,)`. Conditional transitions are callables. Only $\mathbb{P}$ forecasts thesis convergence; the anchor is not a $\mathbb{Q}$ stock price.
O2. Filter / estimate $b_k \in \mathcal{P}(\mathbb{R}_+ \times \mathbb{R} \times \Theta)$: posterior of (v, u, θ). $\widehat{\theta}, \widehat{v}, \widehat{u}$: mode; ℓ_k: measurement likelihood.	Start with a fixed proper prior. Measurements: $\log Y^v \mid v \sim \mathcal{N}(\log v, \sigma_v^2)$, $y \mid u \sim \mathcal{N}(u, \tau_y^2)$, $C_j \mid Z, \theta \sim \mathcal{N}(c_j^A, \omega_j^2)$, conditionally independent; omit missing observations. With the O1 joint transition, $b_{k+1}(v,u,\theta) \propto \ell_{k+1}(v,u,\theta) \iint p_\theta^P(S_{k+1}, v, u \mid S_k, v', u') b_k(v', u', \theta)\,\mathrm{d}v'\,\mathrm{d}u'$. Normalize; use quadrature/particles and update released measurements between decisions. Parameters remain fixed along each future path; decisions integrate the posterior, not just its mode. Mode Greeks are a local constraint approximation. **To O3–O5.**	Weighted particles `(N_particles,14+p)` or quadrature; fit by likelihood/Bayes update. Observation noise $\sigma_v > 0$; retain posterior uncertainty.
O3. Path forecast then reprice P_k^H: physical law on $Z_{k:k+H}$. $c_j^A \in \mathbb{R}_+$: dollars/contract; $G \in \mathbb{R}^{3\times J}$: delta/gamma/vega. $g_j(S) = [\chi_j(S - K_j)]_+$.	$\mathsf{P}_k^H = \int \text{Law}_\theta^P(Z_{k:k+H} \mid S_k, v, u) b_k(\mathrm{d}v, \mathrm{d}u, \mathrm{d}\theta)$. $\mathbb{P}(Z_{k+H} \mid \mathcal{F}_k)$ is only its endpoint marginal. Draw real-world *paths*, catalyst outcomes and future noisy observations. At *each* simulated state $Z_l^{(p)}$, mark each live option with the paired $\mathbb{Q}$ Snell solver: $c_j^A(l, S, v) = \max\{m_j g_j(S), e^{-\int_l^{l'} r\,\mathrm{d}t} \mathbb{E}_\theta^Q[c_j^A(l', S', v') \mid S, v]\}$, terminal $m_j g_j$. Refine the exercise grid (including dividends/expiry) to a fixed tolerance. $G_{\cdot j} = (\partial_S c_j^A, \partial_{SS} c_j^A, 2\sqrt{v}\partial_v c_j^A)^\top$. Thus $\mathbb{P}$ weights future outcomes; $\mathbb{Q}$ gives conditional option marks, never probabilities of thesis success. Re-filter synthetic observations before each future action. **To O5–O7.**	Paths `(P,H+1,3)` (subdivide for diffusion/events); marks `(P,H+1,J)`, Greeks `(3,J)`; price/filter callables.
O4. Thesis mode / instruments $p_k = b_k(d_k > \delta)$; $z_k \in \{0, 1, \dagger\}$: waiting, active, closed; $p_{\text{on}} > p_{\text{off}}$. $E_k \in \{0,1\}^{m\times J}$: dated leg selector; chosen $C \in \mathbb{R}^m$, $G \in \mathbb{R}^{3\times m}$.	Activate at $p \ge p_{\text{on}}$; retain between thresholds. Exit on convergence $\mathbb{E}d \le \delta_{\text{exit}}$, $p \le p_{\text{off}}$, released adverse evidence $y < y_{\min}$ or $c_j < \underline{c}_j$, or fixed T^{th}; freeze thresholds and distinguish reasons. Waiting forbids new thesis risk. Closed † cancels entries and flattens thesis exposure, retaining delivery hedges; it is absorbing until a new record. Select live legs by quotes, tenor, liquidity/spread caps and delivery terms; retain held legs for exit. Use $EC, GE^\top$. Recompare *shares, calls, spreads, hedges and zero*. Expiry must cover holding or have a roll/expiry branch. An active thesis may hold zero. Baseline disables this gate and sets $\kappa_d = 0$. **To O5–O6.**	Probability/mode/deadline scalars; indices `(m,)`, selected arrays. Contract expiry, planning horizon and thesis deadline are distinct.

SPY options: dynamic policy and a multi-week example

The action is an instrument/hedge/exercise decision conditioned on a thesis belief. Assignment and future price paths remain random.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
O5. Baseline and longer control Account $s^{\text{acct}} = (q, h, B, \mathfrak{p})$, $q \in \mathbb{Z}^m, h \in l_h\mathbb{Z}, B \in \mathbb{R}$; $\mathfrak{p}$: pending claims. $a = (\Delta q, \Delta h, e^+)$: bounded integer/lot trades and owned exercise. $s = (b, z, T^{\text{th}} - t, s^{\text{acct}}, \text{pending orders})$.	Baseline $H = 1$: $a^* = \arg\max_{a \in \mathcal{A}(s)} P^{-1} \sum_p U(W_{k+1}^{\text{liq},(p)}(a))$, $U(w) = w - \lambda_U\sqrt{a_U^2 + w^2}$, $a_U > 0, 0 < \lambda_U < 1$. The ledger below generates each wealth sample. For $H > 1$, $Q_k^h(s,a) = \mathbb{E}_k^P[-\eta_T\|D_T a_{\text{trade}}\|_1 + V_{k+1}^{h-1}(s')]$, $V_k^h(s) = \max_{a \in \mathcal{A}(s)}[Q_k^h(s,a) - h_0 \mathbf{1}_{a_{\text{trade}} \neq 0}]$. At the deadline set $V^0 = U(W^{\text{liq}})$; earlier boundaries use this liquidation approximation or an estimated continuation value. One-period baseline has $\eta_T = h_0 = 0$. Expectations also average assignment/fills. Stochastic MPC executes the first action of a nonanticipative tree; full Bellman control plans to the deadline. Future actions see observations/posteriors, not latent states or later shocks. **To O6–O7.**	Finite action enumeration or global optimizer; CVXPY arrays for convex subproblems. D_T is a fixed positive diagonal trade-unit scale; $\eta_T, h_0 \geq 0$ are noncash preferences.
O5a. Risk / no-action region $\mathcal{A}(s)$: feasible controls; $b_G \in \mathbb{R}_+^3$, $\bar{q} \in \mathbb{Z}_+^m$. $\Pi_p = W_p'^{\text{liq}} - R_f W$, $R_f = e^{\int r\,\mathrm{d}t}$; $\alpha \in (0,1)$.	At every node/interim event enforce $\lvert q\rvert \leq \bar{q}$, $0 \leq e^+ \leq (q + \Delta q)_+$, legal exercise dates, $\lvert Gq + (h + h^{\text{pend}}, 0, 0)^\top\rvert \leq b_G$, and available cash minus broker margin $\geq R_{\text{cash}}$, including joint-assignment stresses. One-step tail constraint: $\widehat{\text{ES}}_\alpha(-\Pi) = \min_\zeta\{\zeta + [P(1-\alpha)]^{-1}\sum_p[-\Pi_p - \zeta]_+\} \leq L_{\text{ES}}$. If hold is feasible, the full no-action region is $\max_{a \neq 0}[Q^H(s,a) - h_0\mathbf{1}_{a_{\text{trade}} \neq 0}] \leq Q^H(s,0)$, also comparing owned exercise. Cash costs are already in wealth; turnover/switch penalties add inertia. Mandatory exercise/risk exits and O4 closure override waiting. Use fixed ties. **To O5 feasible actions.**	P&L samples `(P,)` per action; conditional mean/variance by sample moments; ES hinges; record value of holding and best alternative.
O6. Optimal execution is separate $y_e, v_e \in \mathbb{R}^{m+1}$: remaining parent trade and child rate; $T_e \ll \Delta_d$. $\Lambda_e \succ 0, \Omega_e \succeq 0 \in \mathbb{R}^{(m+1)\times(m+1)}$.	Given O5's trade target, minimize $\int_0^{T_e}[s_e^\top\lvert v_e\rvert + v_e^\top\Lambda_e v_e + \lambda_e y_e^\top\Omega_e y_e]\,\mathrm{d}u$, $\dot{y}_e = -v_e$, $y_e(0) = a_{\text{trade}}$, $y_e(T_e) = 0$, with lot/rate/legging/margin limits. Fit impact from fills and Ω_e from short-horizon quote moves. Replace O5's forecast cash costs by expected execution costs; its risk penalty is not cash. Immediate full fills remain baseline. Pending/unfilled legs return to strategic state; owned-exercise instructions bypass this trade scheduler. **To O7 fills/exercise.**	Execution optimizer/callable, arrays `(m+1,)`, order book and fill records. Allocation selects exposure; execution schedules it.
O7. Cash, delivery and assignment $K \geq 0$: actual trade cost; $e = e^+ - A \in \mathbb{Z}^m$. A_j: assigned shorts; $\beta_A, f_j \in \mathbb{R}^5$. $B^{\text{pend}}, h^{\text{pend}} \in \mathbb{R}$: unsettled cash/shares.	For actual fills: $B' = B - C^\top\Delta q - S\Delta h - K$, $q' = q + \Delta q, h' = h + \Delta h$. On exercise, $q'' = q' - e$, $\Delta h^{\text{pend}} = \sum m_j\chi_j e_j$, $\Delta B^{\text{pend}} = -\sum m_j K_j\chi_j e_j$. Transfer at delivery once. Pre-expiry baseline $A_j \sim \text{Binom}((-q_j')_+, \sigma(\beta_A^\top f_j))$, independent conditionally; $\sigma(x) = (1 + e^{-x})^{-1}$, $f = (1, \log(S/K), T - t, C - mg, D_{\text{next}}^{\text{div}})$. Fit bounded β_A by binomial likelihood. At expiry use the declared exercise/contrary-notice rule; remove unexercised legs. Notices are immediate; delivery follows its calendar. Cash accrues as $B \leftarrow R_\ell[B]_+ - R_b[-B]_+ - \text{Borrow} + h^{\text{ent}}D^{\text{div}}$. $W = B + B^{\text{pend}} + (h + h^{\text{pend}})S + q^\top C$. Liquidate at executable costs for W^{liq}; actual future marks replace predicted $\mathbb{Q}$ marks. Baseline $K = s^\top\lvert a_{\text{trade}}\rvert + a_{\text{trade}}^\top\Lambda a_{\text{trade}}$, $\Lambda \succeq 0$: observed spreads/fees and fitted impact, held fixed in forecasts. **To O5 and next posterior/account.**	Event ledger / pending-claim records, assignment samples `(P,H,m)`; cash/borrow/entitlement schedules as supplied functions. Only owned exercise is controlled.
O8. RL and validation $Q_\psi : \mathbb{R}^{d_s+d_a} \to \mathbb{R}$, $d_a = 2m + 1$; $\varphi_s(s) \in \mathbb{R}^{d_s}$: sufficient finite belief/account encoding.	Optional RL replaces O5's value solve: target $Y = -\eta_T\|D_T a_{\text{trade}}\|_1 - h_0\mathbf{1}_{a_{\text{trade}} \neq 0} + \max_{a' \in \mathcal{A}(s')} Q_{\bar{\psi}}(s', a')$, terminal maximum replaced by $U(W^{\text{liq}})$; loss $\mathbb{E}[(Q_\psi(s,a) - Y)^2]$. Useful for expensive repeated solves or learned action-dependent fills/information; unnecessary for tractable identified MPC. Require supported actions and validated simulator/offline data; preserve hard constraints. Seal chronological tests of path/catalyst calibration, $\mathbb{P}/\mathbb{Q}$ errors, assignment, costs, turnover and net benchmark gains. Blocks must preserve horizon overlap and longer dependence; adjust multiplicity. Gate solves only for valid/nonempty problems; otherwise funded exit or stop/default. **To control/update gate.**	Model callable / tensors; training tuples (s, a, r, s'). Audit posterior, gap, thesis mode, held-vs-switch value, instrument and exit reason. RL does not create the thesis forecast.

Worked conceptual example (hypothetical, normalized prices). Daily decisions; intraday observations; deadline day 20; a call expires day 40. Let filtered $u \sim \mathcal{N}(\log 110, 0.04^2)$, $\delta = 0$, $p_{\text{on}} = 0.90, p_{\text{off}} = 0.60$, $\delta_{\text{exit}} = 0.01$. **Entry:** at $S = 100$, $p \approx 0.99$; enter only if the call wins the $\mathbb{P}$-path/$\mathbb{Q}$-mark comparison after costs. **Noise:** an intraday dip with unchanged thesis evidence does not trigger a decision or invalidate it. **Week 2:** at $S = 106$, remaining gap $\log(110/106) = 0.037$, $p \approx 0.82$: retain the thesis. If the after-cost value gain from a smaller call spread (or shares) exceeds h_0, switch; otherwise hold. **Exit:** $S = 109$ gives gap $0.009 < \delta_{\text{exit}}$, so close on convergence. Alternatively a failed catalyst moving the anchor to 102 at $S = 106$ gives $p \approx 0.17$: invalidate. Day 20 closes any remaining thesis risk; zero exposure can be optimal earlier while the thesis remains active. **Composition:** $O \to (X, y) \to (b, \widehat{\theta}, d, z) \to \mathsf{P}_k^H \to \{c^{A,Q}(Z_l^{(p)}), W_l^{(p)}\} \to (Q^H, \mathcal{A}) \to a^{\text{str}} \to \text{fills} \to W' \to b'$.

$$a_k^* = F(O_{\leq k}, s_k^{\text{acct}}; b_k, \widehat{\theta}, \widehat{\beta}_A) = \mathcal{E}_{\text{O6}}\left[\text{Gate}_{\text{O4,O8}}\left(\arg\max_{a \in \mathcal{A}(s_k)}\left\{\mathbb{E}_{\substack{Z \sim \mathsf{P}_k^H,\ C_l = c^{A,Q}(Z_l) \\ s' = \text{Ledger}_{\text{O7}}(s_k, a)}}[-\eta_T\|D_T a_{\text{trade}}\|_1 + V_{k+1}^{H-1}(s')] - h_0\mathbf{1}_{a_{\text{trade}} \neq 0}\right\}\right)\right].$$